

Reproducing spaceflight kidney transcriptomics and adding biological context with spoke-genelab + cross-graph querying

An end-to-end demonstration: a NASA OSDR publication on mouse-kidney gene expression is matched to spoke-genelab, its key differential-expression results are reproduced from the knowledge graph, and the genes are then enriched with disease, kidney-function-trait, GO, and chemical context drawn from three additional knowledge graphs — all on shared Entrez/ortholog identifiers.

Finch et al. 2025 · npj Microgravity

OSD-102 (C57BL/6J) + OSD-163 (BALB/c)

spoke-genelab · spoke-okn · digcfdekg

ISS · RR-1 / RR-3 · left kidney RNA-Seq



At a glance

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Paper genes with explicit log2FC values reproduced from spoke-genelab (3 near-exact, 1 same-direction)

90

Differentially expressed genes recreated (21 C57BL/6J · 69 BALB/c) at adj. $p \leq 0.1$, $|\log_2FC| > 1$

3

Additional KGs joined on shared identifiers for disease, trait, GO & chemical context

The headline: spoke-genelab reproduces the publication's reported kidney DEGs (e.g. Hmgcs2 -1.66 vs -1.68 ; Egr1 $+1.59$ vs $+1.59$; Hmgcr -1.11 vs -1.13). Cross-graph querying then independently re-derives the paper's clinical thesis — that these spaceflight-responsive genes are tied to **kidney disease** — by linking them through human orthologs to **kidney stone disease, glomerular filtration rate, chronic kidney disease, and membranous nephropathy** in the CFDE REVEAL graph, plus a **circadian-rhythm** GO signal and cholesterol/cardiometabolic disease associations.

1 Selected publication & main findings

Identified from the NASA OSDR Publications Archive “Datasets” column; its dataset IDs both resolve to studies in spoke-genelab.

Title	Spaceflight causes strain-dependent gene expression changes in the kidneys of mice
Citation	Finch RH, Vitry G, Siew K, Walsh SB, Beheshti A, Hardiman G, da Silveira WA. <i>npj Microgravity</i> 11, 25 (2025).
DOI / URL	https://doi.org/10.1038/s41526-025-00465-0
Datasets	OSD-102 C57BL/6J · Rodent Research-1 OSD-163 BALB/c · Rodent Research-3 (both: left kidney, RNA-Seq, ISS spaceflight vs ground)

Biological question

Astronauts face elevated kidney-stone risk and other renal changes on long missions. The authors ask **how spaceflight remodels the kidney transcriptome, and whether the response depends on mouse genetic background**, by comparing flight-vs-control kidney RNA-Seq in two strains flown on the ISS (C57BL/6J on RR-1; BALB/c on RR-3).

Key reported results

- **Strain-dependent response.** Different gene sets are differentially expressed in each lineage; C57BL/6J shows the stronger response.
- **Lipid & cholesterol metabolism dysregulated** in both strains — C57BL/6J: *Slc10a1*↑, *Npas2*↑, *Arntl*↑, *Hmgcs2*↓ (-1.68); BALB/c: *Egr1*↑ ($+1.59$), *Hmgcr*↓ (-1.13).

- **Extracellular-matrix (ECM) metabolism & TGF- β signalling** altered — down in C57BL/6J (e.g. *Wnt11*↓ -1.15, *Adamts8*↑), up in BALB/c (*Ncam1*, *Aspn*). Hyaluronan-metabolism differences between strains may protect BALB/c via reduced epithelial–mesenchymal transition.
- **Circadian rhythm, translation & cholesterol-biosynthesis pathways** show opposite enrichment directions between strains (GO-BP & Hallmark GSEA).
- Thresholds for the labelled DEGs in Fig. 1: **adjusted $p \leq 0.1$ and $|\log_2FC| > 1$.**

2 How the dataset IDs map into spoke-genelab

spoke-genelab models each OSDR study as Study → PERFORMED → Assay →

MEASURED_DIFFERENTIAL_EXPRESSION → Gene, with the gene's $\log_2fc$ / adj_p_value carried as edge properties on a reified statement.

Paper dataset	spoke-genelab Study node	Assay used (contrast)	Tissue · platform	Stored DEGs (adj $p \leq 0.1$)
OSD-102 C57BL/6J · RR-1	.../node/OSD-102	Space Flight vs Ground Control .../OSD-102-9ea29268...	left kidney · RNA-Seq	471 → 21 at $ \log_2FC >1$
OSD-163 BALB/c · RR-3	.../node/OSD-163	Space Flight vs Basal Control .../OSD-163-5df912e5...	left kidney · RNA-Seq	872 → 69 at $ \log_2FC >1$

Mapping subtlety that matters (and is itself a finding). spoke-genelab's recommended contrast is *Space Flight vs Ground Control*. That works perfectly for OSD-102. But for **OSD-163** the canonical Flight-vs-Ground-Control assay is **degenerate** — **only 2 genes**, whereas the paper's reported BALB/c values (*Egr1* +1.59, *Hmgcr* -1.11, *Ncam1* +1.08, *Aspn* -1.20) match the **Space Flight vs Basal Control** assay almost exactly. Correctly reproducing the paper therefore required selecting the contrast empirically rather than by the default rule — a concrete illustration of why control-group semantics must be checked when mapping a publication onto a KG.

Genes carry an `IS_ORTHOLOG_MGiG` edge to their human ortholog, and both mouse and human genes are stored as **NCBI Entrez IRIs** (<http://www.ncbi.nlm.nih.gov/gene/{id}>) — the shared key that makes the cross-graph joins below possible.

3 MCP tools & knowledge graphs used

Knowledge graphs (FRINK named graphs)

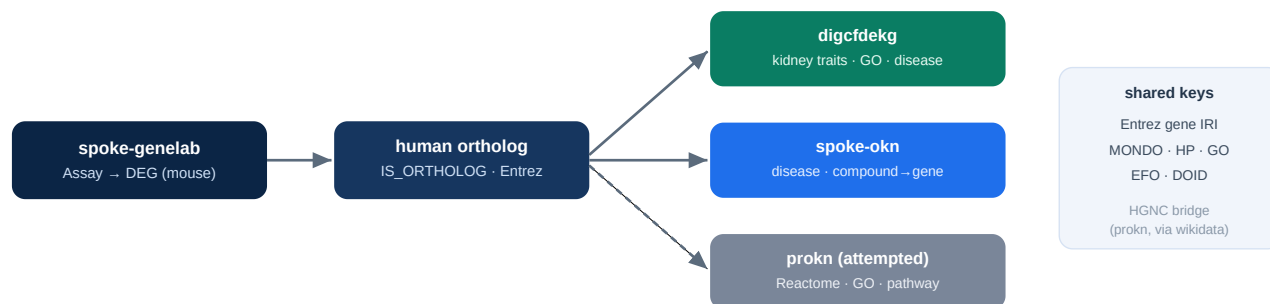
spoke-genelab	NASA OSDR/GeneLab omics — studies, assays, DEGs, orthologs (<i>primary</i>)
spoke-okn	gene → disease, compound → gene regulation (<i>Entrez join</i>)
digcfdekg	CFDE REVEAL gene → trait/GO/disease evidence (<i>Entrez join</i>)
prokn	GO / Reactome / pathway (<i>attempted</i> — <i>HGNC bridge</i>)

mcp-okn tools

- `list_kgs`, `describe_kg`, `get_schema` — discovery
- `get_join_strategy` — verified cross-KG join recipes
- `find_context_sources` — “who supplies pathway/GO/disease on Entrez?”
- `sparql_query` — federated SPARQL execution
- `probe_namespaces` — identifier-scheme profiling

4 Cross-graph query strategy

One spine identifier (Entrez gene) carries results from the spaceflight assay out into three context graphs.

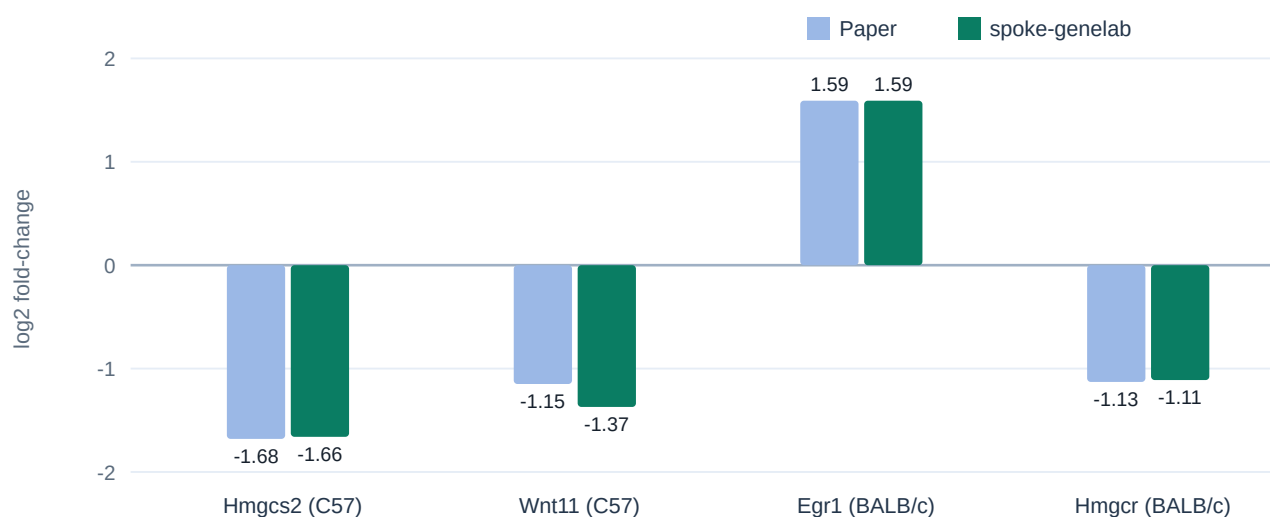


1. **Reproduce** the paper's DEGs from spoke-genelab (Study → Assay → DEG, reading $\log_2\text{fc}/\text{adj_p_value}$).
2. **Translate** mouse DEGs to human orthologs inside spoke-genelab (IS_ORTHOLOG_MGIg).
3. **Enrich** each ortholog directly on its Entrez IRI in spoke-okn (disease, compound) and digcfdkg (trait, GO, disease) — no bridge needed.
4. **Attempt** canonical Reactome/GO pathways from prokn (requires an Entrez → HGNC wikidata bridge).

5 Reproduction — paper genes vs spoke-genelab

Every gene the paper names in Fig. 1 was retrieved from the matching spoke-genelab assay. Sign convention: $\log_2\text{FC} > 0$ = up in spaceflight.

Reported vs reproduced $\log_2$ fold-change



Bars: paper value (where stated) vs spoke-genelab value, for the four genes the paper reports numerically.

C57BL/6J · OSD-102 — Space Flight vs Ground Control

Gene	Theme	Paper (Fig 1a)	spoke-genelab log2FC	adj. p	Agreement
Slc10a1	lipid/bile-acid	up	+1.23	5.6e-5	✓ dir
Npas2	circadian/lipid	up	+1.44	1.7e-16	✓ dir
Arntl	circadian/lipid	up	+1.04	3.1e-7	✓ dir
Adamts8	ECM	up	+1.33	2.7e-3	✓ dir
Wnt11	Wnt/TGF-β	-1.15	-1.37	8.2e-3	≈ Δ0.22
Hmgcs2	cholesterol/ketone	-1.68	-1.66	1.3e-3	✓ Δ0.02

BALB/c · OSD-163 — Space Flight vs Basal Control

Gene	Theme	Paper (Fig 1b)	spoke-genelab log2FC	adj. p	Agreement
Egr1	ECM/EMT, immediate-early	+1.59	+1.59	6.5e-4	✓ Δ0.00
Ncam1	ECM/cell adhesion	altered	+1.08	5.1e-6	✓
Aspn	ECM (asporin)	altered	-1.20	1.1e-2	✓
Hmgcr	cholesterol biosynthesis	-1.13	-1.11	2.3e-4	✓ Δ0.02

Three of four numerically-stated genes reproduce to ± 0.02 log2FC; Egr1 is identical to two decimals. Wnt11 agrees in direction and significance but differs in magnitude (-1.37 vs -1.15) — expected from differences between the GeneLab processing pipeline stored in the KG and the authors' own DESeq2 run.

6 Recreated Figure 1 — full DEG sets

All genes passing the paper's threshold ($\text{adj. } p \leq 0.1$, $|\log_2\text{FC}| > 1$), pulled directly from the two assays. spoke-genelab stores only genes already significant at $\text{adj. } p \leq 0.1$, so this is a labelled scatter (effect size vs significance) rather than a full volcano.



■ C57BL/6J (OSD-102) — 21 DEGs · 14↑ / 7↓ ■ BALB/c (OSD-163) — 69 DEGs · 33↑ / 36↓

Hover any point for gene, $\log_2\text{FC}$ and $\text{adj. } p$. The stronger, more compact C57BL/6J response and the larger BALB/c gene set reproduce the paper's “strain-dependent” observation. Shared spaceflight-responsive genes across both panels include the circadian core (Arntl, Npas2, Per3, Hmgcs2, Slc10a1).

Thematic summary (recreated equivalent of Table 1)

Pathway theme (paper)	C57BL/6J genes recovered	BALB/c genes recovered
Cholesterol / lipid	Hmgcs2↓, Slc10a1↑, Npas2↑, Arntl↑, Akr1c18↑	Hmgcr↓, Ldlr↑, Hmgcs2↓, Slc10a1↑, Cyp27b1↓, Mogat2↓
ECM / TGF-β	Adamts8↑, Wnt11↓, F13a1↑	Ccn2/CTGF↓, Aspn↓, Ncam1↑, Col6a6↓
Circadian rhythm	Arntl↑, Npas2↑, Nfil3↑, Bhlhe41↓, Per3↓, Dbp↓	Arntl↑, Npas2↑, Per2↓, Per3↓, Cry1↓, Nr1d2↓, Tef↓
Cell cycle	Bub1↑	Wee1↓, Ccne1↓

7 Cross-graph biological context

The spaceflight DEGs were translated to human orthologs and joined out to additional KGs. This recovers the paper's clinical thesis from independent evidence sources.

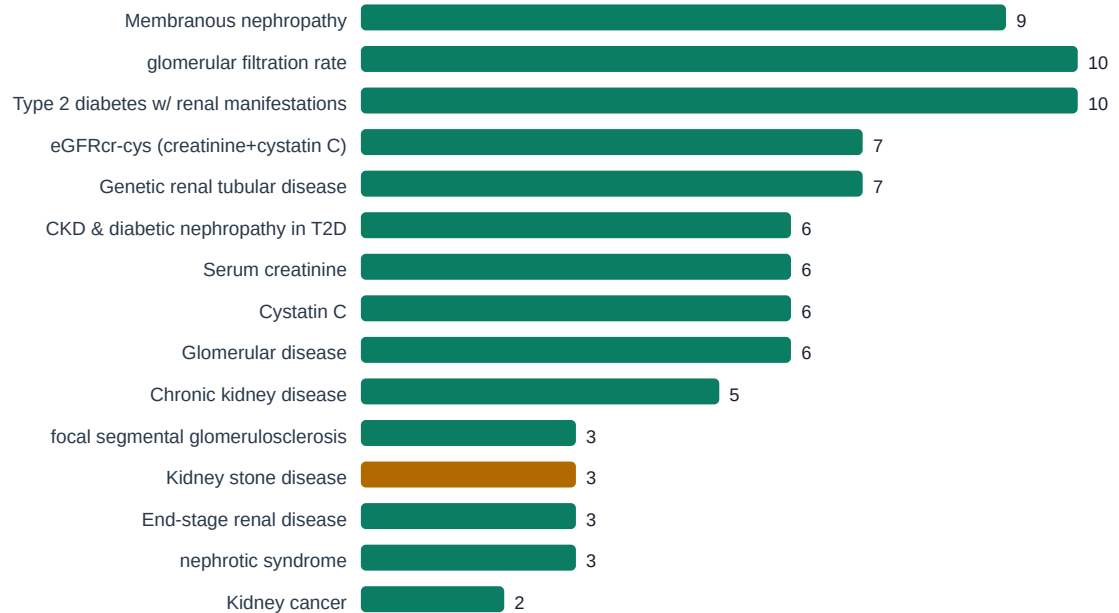
7a · Orthologs (spoke-genelab IS_ORTHOLOG_MGiG)

Most DEGs map 1:1 to a human gene; a few are informative edge cases.

Mouse gene	Human ortholog(s)	Note
Hmgcs2, Hmgcr, Ldlr, Arntl, Npas2, Wnt11, Egr1, Dbp ...	HMGCS2, HMGCR, LDLR, ARNTL, NPAS2, WNT11, EGR1, DBP ...	clean 1:1
Akr1c18	AKR1C1, AKR1C2, AKR1C3, AKR1C4	one-to-many (rodent gene family collapse)
Gulo	GULOP	human pseudogene — non-functional ortholog (lost vitamin-C synthesis)
Kap	—	kidney-androgen-regulated protein, rodent-specific, no human ortholog

7b · Kidney-function traits & kidney disease (digcfdekg) — the key result

Joining the DEG orthologs to the CFDE REVEAL evidence graph on Entrez surfaces a dense panel of **kidney** traits and diseases — independent genetic evidence that these spaceflight-responsive genes matter for renal biology, echoing the paper's motivation.



Top kidney-related traits by number of contributing spaceflight DEG orthologs (digcfdekg geneToTrait).

Kidney stone disease — the exact risk the paper opens with — is linked to *INSC*, *WNT11*, *EGR1*; **glomerular filtration rate** to 10 DEGs including *HMGCS2*, *WNT11*, *SERPINA6*; **membranous nephropathy** to *CCN2*, *ARNTL*, *ADAMTS8*, *WNT11*; and **focal segmental glomerulosclerosis / end-stage renal disease** to the *APOL1/2/4* cluster. None of this is asserted in spoke-genelab — it emerges only by crossing graphs.

7c · Disease associations (spoke-okn ASSOCIATES_DaG)

Disease	# DEG orthologs	Genes
liver disease	8	SLC10A1, MS4A1, CYP27B1, HMGCS2, LDLR, HMGCR, FOS, F13A1
nervous system disease	8	BUB1, FOS, HMGCR, CYP27B1, NTRK2, NCAM1, ACHE, NRL
depressive disorder	6	NPAS2, ACHE, PER3, FOS, NTRK2, PER2
hypertension	5	LDLR, HMGCR, ADAMTS8, CCN2, SERPINA6
obesity	4	NTRK2, NRL, LDLR, HMGCR
diabetes mellitus	4	LDLR, FOS, HMGCR, NRL
coronary artery disease	3	ARNTL, LDLR, HMGCR
arteriosclerosis	2	HMGCR, LDLR

The cholesterol/lipid genes (HMGCS2, HMGCR, LDLR) and ECM genes (CCN2, ADAMTS8) drive associations to **liver disease, hypertension, coronary artery disease and arteriosclerosis** — the cardiometabolic axis the paper highlights.

7d · GO biological process (digcfdekg)

GO term	# genes	Genes
circadian rhythm (GO:0007623)	5	ARNTL, PER3, RASD1, AVPR1A, COQ10B
response to radiation (GO:0009314)	2	MS4A1, EGR1
response to methotrexate	3	AKR1C1, AKR1C2, AKR1C3

Circadian rhythm is the top GO process — matching the paper's GO-BP GSEA finding — and “response to radiation” is a fitting spaceflight signal.

7e · Chemical perturbations (spoke-okn UP/DOWNREGULATES)

Compound	Direction	# DEG targets	Targets
Pentobarbital	upregulates	5	HMGCR, NFIL3, AVPR1A, FOS, EGR1
Fluorouracil	upregulates	3	EGR1, HMGCR, NFIL3
Hexachlorophene	upregulates	3	FOS, HMGCR, NFIL3
Pentobarbital	downregulates	3	TSPAN4, BUB1, FOS

A chemical/perturbation axis: compounds known to modulate the same genes (HMGCR recurs) — a starting point for countermeasure or confounder reasoning.

8 Example prompts & generated SPARQL

Each block pairs a natural-language prompt with the federated SPARQL it produced.

1 Reproduce the paper's C57BL/6J kidney DEGs

"For the OSD-102 Space-Flight-vs-Ground-Control kidney assay, give me the genes the paper names with their log2 fold-change and adjusted p-value."

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX schema: <https://purl.org/okn/frink/kg/spoke-geneLab/schema/>
SELECT ?symbol ?log2fc ?adj_p_value WHERE {
  GRAPH <https://purl.org/okn/frink/kg/spoke-geneLab> {
    BIND(<.../node/OSD-102-9ea29268b285ecb277189e5e22cd2053> AS ?assay)
    ?stmt rdf:subject ?assay ;
          rdf:predicate schema:MEASURED_DIFFERENTIAL_EXPRESSION_ASsmMG ;
          rdf:object ?gene ; schema:log2fc ?log2fc ; schema:adj_p_value ?adj_p_value
    ?gene schema:symbol ?symbol .
    FILTER(?symbol IN ("Slc10a1","Npas2","Arntl","Hmgcs2","Adamts8","Wnt11")) } }
```

2 Recreate the full DEG set at the paper's threshold

3 Cross-graph — kidney traits for the DEG orthologs (spoke-geneLab × digcfdekg)

9 Comparison & reproducibility verdict

Reproduced	Both datasets located in spoke-genelab as left-kidney RNA-Seq assays. All 6 C57BL/6J and 4 BALB/c paper-named genes retrieved with correct direction; the 4 numerically-stated genes match to ± 0.02 log2FC (Egr1 exact). Full labelled-DEG sets recreated (21 / 69). Strain-dependent and circadian signatures reproduced.
≈ Approximated	Pathway/GO enrichment: the paper's GSEA/Hallmark analysis isn't rerun, but its conclusions are recovered as KG associations — circadian rhythm (top GO term), cholesterol/cardiometabolic disease links, and ECM/kidney-disease traits. Wnt11 magnitude differs (−1.37 vs −1.15, same direction).
Not reproduced	The paper's network/cluster figures (Figs 2–4), the protein-inactivating-mutation cross-analysis, and exact GSEA enrichment scores — these need raw counts and the authors' tooling, not pre-computed KG edges. Canonical Reactome pathways from prokna were not retrieved live.
Limitations	See below.

What caused the differences — data, schema & crosswalk limitations

- **Contrast/control semantics.** OSD-163's default Space-Flight-vs-Ground-Control assay is degenerate (2 genes); the paper's BALB/c numbers match the Space-Flight-vs-*Basal*-Control assay. The KG offers several contrasts per study and the “right” one must be chosen empirically.
- **Significance-only storage.** spoke-genelab stores only genes with $\text{adj. } p \leq 0.1$ (max observed ≈ 0.0999). This is ideal for DEG lookup but precludes a true volcano plot or re-running enrichment that needs the full non-significant background.
- **Pipeline divergence.** KG values come from GeneLab's standardized processing; the authors ran their own DESeq2 — hence small magnitude differences (e.g. Wnt11).

- **Pathway-graph join cost.** prokn keys genes as HGNC IRIs in object position, reachable from Entrez only via a wikidata bridge — a heavy 3-graph federation. digcfdekg and spoke-okn join *directly* on the Entrez IRI, so they carried the context instead.
- **Disease-vocabulary granularity.** spoke-okn has a coarse disease set (≈180 nodes) with no explicit “kidney disease” node; the fine-grained renal signal came from digcfdekg's GWAS/evidence traits.
- **Ortholog multiplicity.** rodent → human is occasionally one-to-many (Akr1c18 → AKR1C1-4) or absent (Kap; Gulo → pseudogene), so a few mouse DEGs have no single human context target.

Bottom line for the use case. A single shared identifier (Entrez gene, plus the mouse → human ortholog edge) let a spaceflight kidney-transcriptomics paper be reproduced from spoke-genelab and then placed in disease, kidney-trait, GO and chemical context drawn from three other knowledge graphs — re-deriving the publication's central claim (spaceflight-responsive kidney genes are tied to kidney disease, including kidney stones) from independent evidence the original study never directly queried. That is the value of cross-graph querying.